Design and Development of IoT-Based Smart Maldive Fish Processing System

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The Maldive fish processing industry plays a significant role in the seafood sector of Sri Lanka and surrounding regions. However, the final-stage quality evaluation of dried fish-particularly for species such as mackerel, yellowfin tuna, and skipjack tuna-remains largely dependent on manual inspection and subjective judgment. Operators visually assess cracks, discoloration, texture irregularities, and odor variations to categorize products into Grade A, Grade B, or Reject classifications. This manual process often leads to inconsistent grading, increased spoilage risk, and reduced export reliability.

This research proposes an AI-Based Multi-Sensor Final Quality Evaluation System designed to automate post-drying inspection. The system integrates computer vision techniques for surface defect detection with moisture and volatile organic compound (VOC) sensors to quantify residual moisture and odor variations. A multi-sensor data fusion framework combines visual and physical parameters to improve grading accuracy and reliability. The system classifies dried fish batches into standardized Grade A, Grade B, and Reject categories and provides real-time monitoring through a dashboard interface.

The proposed solution aims to improve grading consistency, reduce subjective bias, and enhance product quality stability. By integrating IoT-based sensing and artificial intelligence within a unified evaluation framework, the system contributes to smart industrial automation and sustainable seafood processing.

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TABLE OF CONTENT

DECLARATION....2
LIST OF TABLES....5
LIST OF FIGURES....6
LIST OF ABBREVIATIONS....6
1. INTRODUCTION....7
1.1 Background and Context....7
1.2 Literature Review....9
1.3 Research Gap....11
2. OBJECTIVE....13
2.1 Main Objective....13
2.2 Specific Objectives....13
3. METHODOLOGY....14
4. HIGH-LEVEL SYSTEM ARCHITECTURE....18
5. USER REQUIREMENTS....21
5.1 Stakeholder Identification....21
6. COMMERCIALIZATION PLAN....23
7. BUDGET AND JUSTIFICATION....25
8. WORK BREAKDOWN STRUCTURE (WBS)....26
9. GANTT CHART....26
10. REFERENCE LIST....27
11. APPENDICES....28

LIST OF TABLES

Table 1-abbreviations....6
Table 2 - Summary of research gaps....12
Table 3- Tools....16
Table 4 - Budget and Justification Table....25

LIST OF FIGURES

Figure 1 – High level System Diagram.....18
Figure 2 - Work Breakdown Structure.....26
Figure 3 - Gant Chart.....26

LIST OF ABBREVIATIONS

Table 1-abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
CNN	Convolutional Neural Network
VOC	Volatile Organic Compound
ML	Machine Learning
DSR	Design Science Research
DFD	Data Flow Diagram
API	Application Programming Interface
UI	User Interface
UX	User Experience
TLS	Transport Layer Security
SSL	Secure Socket Layer
JWT	JSON Web Token
GPU	Graphics Processing Unit
ROI	Return on Investment

1. INTRODUCTION

1.1 Background and Context

Real-World Problem

Maldive fish processing is an important traditional seafood industry in Sri Lanka and the Maldives, where species such as mackerel (Rastrelliger spp.), yellowfin tuna (Thunnus albacares), and skipjack tuna (Katsuwonus pelamis) are boiled, salted, smoked, and dried to produce preserved products with extended shelf life. In a fully automated smart processing environment, the final stage after drying plays a crucial role in determining whether the product is suitable for packaging and market distribution. However, this stage remains largely dependent on manual inspection and subjective evaluation.

Operators visually inspect dried fish, assess surface texture through touch, and detect odor changes through experience to categorize products into Grade A (premium quality), Grade B (acceptable quality), and Reject (defective or unstable quality). Due to variations in drying temperature, humidity, airflow, and fish characteristics, inconsistencies arise in moisture levels, surface texture, discoloration, and structural integrity. Over-drying leads to excessive brittleness and cracks, while under-drying increases moisture retention and microbial risk. Uneven drying patterns produce inconsistent texture and color, especially across different fish categories such as mackerel, yellowfin tuna, and skipjack tuna, which vary in fat content and muscle structure. The absence of standardized, measurable grading mechanisms results in inconsistent quality decisions and potential spoilage.

Who Is Affected and Why It Is Important

This issue affects multiple stakeholders across the seafood value chain. Small and medium-scale processors experience economic losses due to inconsistent grading and post-processing spoilage. Exporters face challenges in meeting international quality and safety standards when grading is not standardized. Consumers are impacted by variability in texture, odor, and product safety. In export-driven economies, seafood quality inconsistency reduces market competitiveness and long-term sustainability.

Accurate and objective final-stage evaluation is therefore essential to ensure product safety, maintain consumer trust, and enhance export reliability. Automating this stage improves grading consistency and supports data-driven packaging decisions, which are critical in maintaining stable shelf life and reducing economic losses.

Statistical Evidence (Local/Global)

Globally, the Food and Agriculture Organization (FAO) reports that approximately 14% of food production is lost between harvest and retail, with seafood being particularly vulnerable to

spoilage due to moisture instability and inadequate post-processing control [1]. Dried seafood products are highly sensitive to residual moisture levels, which directly influence microbial growth and oxidative degradation [2]. Research shows that improper moisture control significantly shortens shelf life and increases spoilage probability in dried fish products [3].

In South Asian dried fish industries, traditional processing methods often lack standardized moisture measurement and automated grading systems, leading to inconsistent product quality and elevated rejection rates in export markets [4]. These findings highlight the need for intelligent evaluation systems specifically tailored for dried seafood products.

Analysis of Existing Solutions and Their Limitations

Modern food-processing industries have adopted computer vision and artificial intelligence (AI) to improve automated inspection. Deep learning techniques, particularly Convolutional Neural Networks (CNNs) and object detection models such as YOLO, have demonstrated high accuracy in detecting surface defects in agricultural and seafood products [5], [6]. AI-based seafood inspection systems have improved consistency in freshness detection and defect identification [7].

However, existing systems present several limitations:

- Most focus on fresh fish freshness detection, not post-drying quality evaluation.
- Visual-only systems ignore critical parameters such as moisture percentage and odor (VOC levels).
- Multi-sensor fusion approaches are rarely applied to dried seafood products [8].
- No documented system classifies dried fish into standardized Grade A, Grade B, and Reject categories using combined moisture, texture, odor, and visual defect analysis.
- Industry-specific optimization for species such as mackerel, yellowfin tuna, and skipjack tuna is largely absent.

Proposed Solution and Its Components

To address these limitations, this research proposes an AI-Based Final Quality Evaluation and Intelligent Grading System integrated within a fully automated processing framework. The system operates immediately after the drying stage and includes the following components:

1. Object Detection Models (e.g., YOLO) to detect cracks, discoloration, and uneven texture.
2. Moisture Sensors to quantify residual moisture percentage.
3. VOC Sensors to evaluate odor variations associated with spoilage.
4. Multi-Sensor Data Fusion Algorithms to combine visual and physical measurements.

5. Machine Learning Classification Models to categorize batches into Grade A, Grade B, and Reject.

The system identifies specific final-stage conditions such as over-drying, under-drying, discoloration, and uneven texture, enabling objective and reproducible grading decisions. This intelligent framework improves packaging suitability determination and reduces subjectivity in quality assessment.

Relevant SDG Alignment

The proposed system contributes to multiple United Nations Sustainable Development Goals (SDGs):

- SDG 2 – Zero Hunger: Reducing post-processing food losses [1].
- SDG 9 – Industry, Innovation and Infrastructure: Introducing AI-driven automation in traditional industries [7].
- SDG 12 – Responsible Consumption and Production: Minimizing waste through accurate grading and packaging optimization.

Research Cluster Alignment

This research falls under the **AIMS – Autonomous Intelligent Machines and Systems** research cluster. The proposed system integrates intelligent perception (computer vision-based defect detection), sensor-driven environmental awareness (moisture and VOC sensing), multi-sensor data fusion, and machine learning-based classification. By enabling autonomous evaluation and grading without human intervention, the system reflects the core objective of the AIMS cluster—developing intelligent machines capable of perceiving, reasoning, and acting within complex industrial environments.

1.2 Literature Review

Recent research in food quality evaluation and processing automation has increasingly focused on integrating artificial intelligence (AI), computer vision, and multi-sensor technologies to improve consistency, reduce human subjectivity, and enhance product grading. Several studies have explored digital monitoring and defect detection in food products, demonstrating that AI and machine learning methods can achieve high accuracy and repeatability in tasks traditionally performed manually.

Computer vision and deep learning have been widely applied in quality inspection of agricultural and seafood products. Oliveira *et al.* developed deep learning-based defect detection models using convolutional neural networks (CNNs) for automated recognition of

visual defects in processed foods, achieving high classification performance in controlled environments [1]. Similar work by Redmon *et al.* introduced real-time object detection frameworks (e.g., YOLO) that enable rapid identification of multiple surface defects, which has been influential in industrial inspection systems [2]. In seafood quality research, Lin *et al.* investigated machine vision methods for non-destructive freshness evaluation, demonstrating that visual texture and color features can indicate spoilage in fresh fish [3]. However, these approaches predominantly focus on fresh fish quality and do not specifically address post-drying evaluation for dried fish products like Maldives fish.

In parallel, scholars have studied sensor-based monitoring systems to track environmental and product parameters. Singh *et al.* developed moisture stability and shelf-life prediction models for dried food products by analyzing moisture content and physicochemical properties, highlighting the importance of quantitative measurements in determining product stability [4]. Similarly, Wang and Liu reviewed multi-sensor data fusion techniques for food quality prediction, showing that combining heterogeneous data sources (e.g., moisture, temperature, visual features) enhances predictive accuracy compared to single-sensor approaches [5]. These studies indicate the advantage of multi-parameter evaluation but stop short of implementing integrated grading systems with concrete quality categories such as Grade A, Grade B, and Reject.

The application of the Internet of Things (IoT) and AI in seafood processing has been reviewed by Yang *et al.*, who found that AI-assisted monitoring systems improve consistency, reduce processing variability, and enable predictive maintenance in processing facilities [6]. Ahmed and Rahman investigated intelligent packaging systems that use sensor feedback and environmental conditions to extend shelf life in seafood and perishable products, illustrating the potential for automated packaging optimization to reduce spoilage [7]. However, existing intelligent packaging solutions typically rely on static rule sets rather than adaptive AI models tailored to the real-time quality conditions of individual batches.

Commercial products and services have also emerged in related domains. Several companies provide automated vision inspection systems for food manufacturing lines that identify defects and sort products based on size or surface features. For example, vision-guided robotic sorters are used in fruit and vegetable grading and in fish fillet inspection. These systems use pre-defined thresholds to classify visual features but generally lack integration with other sensor inputs such as moisture or odor. Likewise, smart drying equipment equipped with environmental sensors can regulate temperature and humidity, but these systems do not perform post-drying quality evaluation or produce packaging recommendations based on quality metrics.

Despite these advancements, significant limitations remain. First, existing research tends to focus on either visual inspection or sensory monitoring, not the fusion of both for multi-factor quality assessment. Second, most automated inspection systems are designed for fresh or raw products, and their algorithms do not generalize well to dried seafood products where surface appearance, moisture levels, and odor interact in complex ways. Third, current commercial solutions rarely implement adaptive AI-based grading into discrete categories (e.g., Grade A,

Grade B, Reject) that reflect both quality and packaging suitability. Finally, packaging recommendation engines in food processing mostly use static decision logic, lacking the capability to adapt recommendations based on real-time multi-sensor data.

These limitations create a clear research gap for a unified system that integrates visual defect detection, moisture and odor analysis, and AI-based decision logic to classify dried Maldive fish into standardized quality grades and provide intelligent packaging recommendations. Such a system would combine the strengths of computer vision, multi-sensor fusion, and predictive analytics, addressing both the inconsistency in manual inspection and the lack of integrated intelligent solutions for dried seafood quality evaluation.

1.3 Research Gap

Despite significant advancements in artificial intelligence, computer vision, and IoT-based monitoring systems for food processing, there remains a substantial gap in the application of these technologies to the final-stage evaluation of dried seafood products such as Maldive fish. Existing research in seafood quality assessment primarily focuses on freshness detection of raw fish or generalized food defect recognition using vision-based techniques [1], [2]. For instance, machine vision models have been successfully developed for detecting freshness indicators in fresh fish fillets using texture and color features [3], while deep learning architectures have shown strong performance in identifying surface defects in agricultural and food products [1]. However, these studies do not address the specific challenges of post-drying quality evaluation in dried fish products, where moisture variability, odor changes, and structural deformation significantly influence shelf life and packaging suitability.

A major limitation of current approaches is their dependence on single-modality inspection systems, particularly vision-only methods [4], [5]. Although object detection models such as YOLO and CNN-based classifiers can effectively identify visible cracks and discoloration, they fail to capture non-visible quality indicators such as residual moisture percentage and volatile organic compound (VOC) levels. Research has demonstrated that moisture stability and water activity are critical determinants of microbial growth and spoilage in dried food products [6]. Multi-sensor fusion approaches have been shown to improve predictive robustness by integrating heterogeneous data sources [7]; however, these techniques are rarely implemented in a unified framework specifically tailored for dried seafood products.

Furthermore, there is no documented system that classifies dried fish into standardized grading categories such as Grade A, Grade B, and Reject based on a combined evaluation of moisture content, surface texture, odor profile, and visual defects. Existing automated inspection solutions often provide binary pass/fail outputs or isolated defect flags without generating structured quality grades aligned with packaging decisions. Additionally, intelligent packaging systems explored in prior studies rely primarily on static decision rules rather than adaptive AI-driven recommendations based on real-time multi-parameter quality assessment [8]. This lack of integration between grading and packaging logic represents a significant technological gap in the dried fish processing industry.

Therefore, the current literature indicates that while AI-based inspection and sensor technologies have matured individually, they remain insufficient when applied independently and have not been integrated into a comprehensive final-stage grading and packaging decision framework for Maldive fish processing.

Table 2 - Summary of research gaps

Feature	[1]	[2]	[3]	[4]	[5]	[6]	[7]	Proposed
Vision-based detection	✓	✓	✓	✓	✗	✓	✓	✓
Moisture measurement	✗	✗	✓	✓	✓	✗	✗	✓
VOC detection	✗	✗	✗	✗	✗	✗	✗	✓
Multi-sensor fusion	✗	✓	✓	✗	✗	✓	✗	✓
Designed for dried fish	✗	✗	✗	✗	✗	✗	✗	✓
A/B/Reject grading	✗	✗	✗	✗	✗	✗	✗	✓
Real-time dashboard	✗	✗	✗	✗	✗	✗	✗	✓

Novelty and Contribution

The proposed research addresses the identified gaps by introducing a unified AI-Based Final Quality Evaluation and Intelligent Packaging Recommendation System specifically designed for post-drying Maldive fish assessment. The primary novelty of this work lies in its integration of multi-sensor fusion and deep learning-based object detection within a single intelligent decision-making framework. Unlike existing systems that rely solely on visual inspection, the proposed approach combines moisture sensors, VOC sensors, and object detection algorithms to evaluate cracks, discoloration, uneven texture, and structural irregularities simultaneously.

Another key contribution is the development of a standardized multi-factor grading mechanism that classifies dried fish batches into Grade A (premium quality), Grade B (acceptable quality), and Reject (defective or unstable quality) categories. This grading system is not limited to visual appearance but incorporates quantitative moisture and odor measurements to ensure shelf-life stability and packaging suitability. By explicitly identifying conditions such as over-drying, under-drying, discoloration, and uneven texture, the system transforms subjective manual grading into an objective, reproducible process.

Additionally, the research introduces an adaptive packaging recommendation engine that tailors packaging parameters according to the evaluated quality metrics. Unlike static packaging approaches discussed in prior work [8], this system links grading outputs directly to

intelligent packaging decisions, creating a closed-loop quality assurance mechanism. The fusion of IoT sensing, computer vision, predictive modeling, and packaging optimization represents a novel contribution to smart food-processing automation.

Overall, this study advances the fields of Artificial Intelligence, Computer Vision, IoT-based sensing, and Smart Manufacturing by developing an end-to-end intelligent grading and packaging decision system specifically for dried seafood products. The proposed framework not only fills a clear research gap but also provides a scalable, industry-specific solution capable of improving grading consistency, reducing spoilage losses, and enhancing economic sustainability in the Maldive fish processing sector.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate an AI-based multi-sensor final quality evaluation system for post-drying Maldive fish that accurately classifies batches into Grade A, Grade B, and Reject categories with at least 90% classification accuracy.

2.2 Specific Objectives

1. **To develop a computer vision-based object detection model** capable of identifying surface defects such as cracks, discoloration, and uneven texture in dried mackerel, yellowfin tuna, and skipjack tuna with a minimum detection precision of 85%.
2. **To integrate moisture and VOC sensors** for real-time measurement of residual moisture percentage and odor variation, enabling quantitative assessment of spoilage risk and drying consistency.
3. **To implement a multi-sensor data fusion framework** that combines visual features, moisture data, and odor metrics to improve grading reliability compared to single-modality inspection systems.
4. **To design a machine learning-based grading model** that classifies dried fish batches into Grade A, Grade B, and Reject categories based on multi-factor evaluation with validated performance metrics (accuracy, precision, recall, F1-score).
5. **Develop a real-time dashboard for batch quality visualization.**

3. METHODOLOGY

Research Approach

This study adopts a **Design Science Research (DSR) approach combined with experimental validation**.

The Design Science methodology is appropriate because the research focuses on developing an innovative AI-based artifact (autonomous grading and evaluation system) to solve a real industrial problem—subjective post-drying quality evaluation in Maldive fish processing.

The study involves:

1. Designing the multi-modal evaluation framework
2. Developing the AI models and sensing mechanisms
3. Implementing a prototype system
4. Experimentally validating system performance against manual grading

The experimental phase is used to evaluate grading accuracy, defect detection performance, and system reliability under real processing conditions.

Technical Methods and Algorithms

The proposed system consists of multiple technical components:

1. Computer Vision for Defect Detection

- **Model Type:** Convolutional Neural Networks (CNN)
- **Object Detection Framework:** YOLOv8 (or similar real-time detector)
- **Purpose:** Detect surface defects such as:
    - Cracks
    - Discoloration
    - Uneven texture
    - Structural irregularities

The object detection model will be trained using labeled image datasets of dried Maldive fish samples.

2. Moisture and Odor Measurement

- **Moisture Sensor:** Digital moisture sensor for residual moisture %
- **VOC Sensor:** Gas sensor (e.g., MQ-series) for odor/spoilage detection
- **Output:** Quantified moisture stability and odor variation levels

These measurements help detect:

- Over-drying
- Under-drying
- Early spoilage risk

3. Multi-Sensor Data Fusion

- Feature-level fusion combining:
    - Visual features (from CNN embeddings)
    - Moisture values
    - VOC levels
- Fusion Techniques:
    - Feature concatenation
    - Standardization and normalization
    - Principal Component Analysis

4. Batch-Level Classification Model

- Algorithms:
    - Random Forest
- Output:
    - Grade A
    - Grade B
    - Reject

The best-performing classifier will be selected based on evaluation metrics.

5. Dashboard Development

- Frontend: React
- Database: MongoDB

The dashboard will display:

- Batch Quality Score
- Defect Types Detected
- Moisture Percentage

- VOC Level
- Drying Status (Over/Under/Stable)

Table 3- Tools

Tool/Platform	**Purpose**	**Justification**
Python	Core development	Strong AI/ML ecosystem
PyTorch	Deep learning	Efficient model training
OpenCV	Image processing	Real-time image handling
YOLOv8	Object detection	High speed & accuracy
Scikit-learn	ML models	Easy model comparison
Raspberry Pi 4	Sensor integration	IoT compatibility
MongoDB	Data storage	Structured batch records
React	Dashboard	Real-time visualization
Google Colab	Model training	Efficient computation

Data Collection and Ethical Considerations

Data Collection

Data will be collected from:

- Images of dried mackerel, yellowfin tuna, and skipjack tuna
- Sensor readings (moisture %, VOC levels)
- Manually labeled quality grades for ground truth

Dataset preparation includes:

- Image labeling
- Quality grade annotation
- Sensor-data logging

Ethical Considerations

- No human subject data will be collected.
- Only product quality data will be analyzed.

- Processing facility data will be anonymized.
- Data will be used strictly for research purposes.
- System outputs will support, not replace, final human authority decisions.

Validation Strategy and Performance Metrics

1. Object Detection Evaluation

- Mean Average Precision (mAP)
- Precision
- Recall
- F1-Score

2. Classification Model Evaluation

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

3. Comparative Validation

- Compare AI grading results with manual grading
- Measure agreement rate (%)
- Analyze misclassification cases

4. System-Level Evaluation

- Dashboard response time
- Sensor accuracy consistency
- End-to-end grading latency

Target Performance Goals:

- ≥ 90% grading accuracy
- ≥ 85% defect detection mAP
- Stable performance across multiple batches

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed system is an AI-Based Final Quality Evaluation System designed to operate at the post-drying stage of Maldive fish processing. The system evaluates dried batches of mackerel, yellowfin tuna, and skipjack tuna and classifies them into Grade A, Grade B, or Reject categories based on visual and sensor-based analysis.

The architecture consists of the following major components:

1. Image Acquisition Module – Captures high-resolution images of dried fish batches.
2. Computer Vision Module – Detects cracks, discoloration, and uneven texture using deep learning.
3. Moisture & VOC Detection Module – Measures residual moisture percentage and odor levels.
4. Multi-Sensor Data Fusion Module – Integrates visual and sensor data.
5. AI-Based Grading Engine – Classifies batches into Grade A / Grade B / Reject.
6. Dashboard & Monitoring System – Displays batch results and analytics.
7. Database – Stores historical batch records and analytics data.

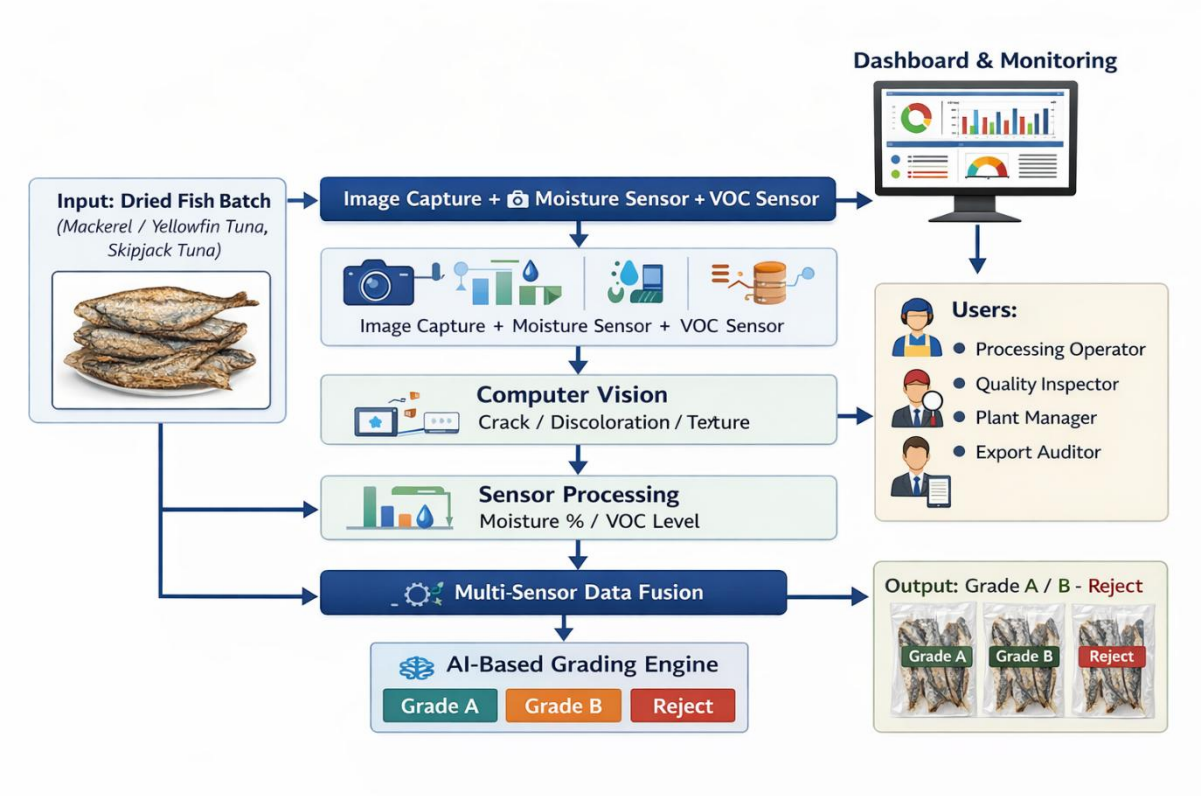


Figure 1 – High level System Diagram

4.2 Component Interaction Flow

This research specifically focuses on the Final Quality Evaluation and Intelligent Packaging Module, which includes:

- Computer Vision Defect Detection
- Moisture & VOC Sensor Integration
- Multi-Sensor Data Fusion
- AI-Based Grading (A/B/Reject)
- Drying Condition Classification
- Real-Time Dashboard

This module serves as the intelligent decision layer of the overall automated system, transforming sensor and image data into structured grading and packaging outputs.

4.3 Component Interactions

1. Image and sensor data are captured.
2. Visual features are extracted using CNN-based object detection.
3. Moisture and VOC values are normalized and processed.
4. Multi-sensor data fusion combines all parameters.
5. The classification model produces Grade A, Grade B, or Reject.
6. Results are displayed on the dashboard and stored in the database.

4.4 Data Flow Explanation

The data flow consists of five main stages:

Data Input

- Image frames of dried fish
- Moisture %
- VOC levels

Preprocessing

- Image resizing and normalization

Feature Extraction

- CNN-based visual feature extraction

- Statistical features from sensor data

Multi-Sensor Fusion & AI Inference

- Classification into Grade A, B, or Reject
- Detection of over-drying / under-drying

Decision & Output

- Quality score generation
- Dashboard visualization

4.6 Scalability Considerations

The system is designed for scalability through:

- Modular Architecture – Additional sensors or AI models can be integrated easily.
- Cloud Deployment – Enables support for multiple processing units.
- Model Retraining Pipeline – Continuous improvement using new data.
- API-Based Communication – Allows integration with larger Industry 4.0 ecosystems.

4.7 Security Considerations

Security measures include:

- Encrypted IoT communication (TLS/SSL)
- Role-based access control for dashboard users
- Secure API authentication (JWT tokens)
- Encrypted database storage

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves multiple stakeholders across the Maldive fish processing ecosystem:

1. Processing Facility Operator

- Responsible for monitoring drying, grading, and packaging stages.
- Use the dashboard to view batch quality scores.

2. Quality Control Inspectors

- Validate AI-based grading decisions (A, B, Reject).

3. Plant Managers / Supervisors

- Monitor production performance and spoilage rates.
- Use analytics for operational decision-making.

4. Exporters / Distributors

- Require consistent grading standards to meet international quality requirements.
- Use system reports for certification and quality assurance.

5. System Administrators

- Manage system configuration, user access, and data storage.
- Ensure secure and stable system operation.

5.2 Functional Requirements

The system must provide the following functional capabilities:

1. **Fish Image Capture** : Capture high-resolution images of dried mackerel, yellowfin tuna, and skipjack tuna batches.
2. **Moisture Measurement** : Measure residual moisture percentage in real time using integrated sensors.
3. **Odor Detection** : Detect VOC levels to identify potential spoilage or abnormal odor.
4. **Defect Detection** : Identify cracks, discoloration, and uneven texture using computer vision models.
5. **Drying Condition Classification** : Classify drying status as over-drying, under-drying, or uneven drying.

6. **Multi-Sensor Data Fusion** : Combine visual and sensor data to generate a unified quality assessment.
7. **Quality Grading** : Classify batches into Grade A, Grade B, or Reject categories.
8. **Real-Time Dashboard Visualization** : Display batch quality scores, defect types, drying conditions, and packaging recommendations.
9. **User Authentication and Role Management :** Provide secure login and role-based access for operators, inspectors, and managers.
10. **System Alerts and Notifications** : Notify users when defective batches or abnormal conditions are detected.

5.3 Non-Functional Requirements

5.3.1 Performance Requirements

- The grading model shall achieve at least **90% classification accuracy**.
- The dashboard shall update in real time without noticeable delay.

5.3.2 Security Requirements

- Data transmission between sensors and server shall be encrypted (TLS/SSL).
- The system shall implement role-based access control.

5.3.3 Usability Requirements

- The dashboard interface shall be simple and intuitive.
- Visual indicators (color-coded grades) shall be used for quick understanding.

5.3.4 Reliability Requirements

- The system shall operate continuously during processing hours.
- Sensor calibration shall be maintained to ensure accurate readings.

5.3.5 Scalability Requirements

- The system shall support multiple processing lines simultaneously.
- Additional sensors and modules shall be easily integrable.

5.3.6 Maintainability Requirements

- AI models shall support periodic retraining with new data.
- System updates shall be deployable without interrupting operations.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Target Market

The primary target market includes:

- Maldive Fish Processing Facilities in Sri Lanka
- Small and Medium Enterprises (SMEs) involved in dried seafood production
- Export-Oriented Seafood Processing Companies
- Fishery Cooperatives and Government Processing Centers
- Food Quality Certification Bodies

Customer Persona

Primary User:

- Role: Processing Plant Manager / Quality Control Supervisor
- Needs: Consistent grading, reduced spoilage, improved packaging decisions
- Pain Points:
    - Inconsistent manual grading
    - Export quality rejections
    - Lack of real-time monitoring

Secondary User:

- Role: Quality Inspector / Operator
- Needs: Easy-to-use dashboard, quick defect detection
- Pain Points: Time-consuming manual inspection , Subjective decision-making

6.2 Value Proposition

The proposed system provides:

- Automated Quality Grading (A/B/Reject) using AI and multi-sensor fusion

- Reduced Spoilage Losses through accurate moisture and defect detection
- Real-Time Monitoring Dashboard for operational insights

Key Value Points:

- Improved compliance with international food safety standards
- Scalable solution for both SMEs and large processing facilities

6.3 Revenue Model

The commercialization strategy will adopt a hybrid revenue model:

1. **System Installation Fee** : One-time cost for hardware setup, sensors, and AI system deployment.
2. **Software Subscription (SaaS Model)** : Monthly or annual subscription for:
    - Dashboard access
    - AI model updates
    - Cloud storage
    - Technical support
3. **Maintenance & Support Services** : Optional yearly maintenance contract for system calibration and upgrades.
4. **Custom Analytics Services** : Premium feature offering production analytics and performance insights.

6.4 Competitive Advantage

The proposed system offers several advantages over existing solutions:

- First integrated system for post-drying Maldive fish evaluation
- Combines Computer Vision + Moisture + Odor Sensors
- Provides Automated Grading (A/B/Reject) rather than simple pass/fail
- Real-time monitoring and analytics dashboard
- Designed specifically for mackerel, yellowfin tuna, and skipjack tuna

6.5 Intellectual Property Considerations

The project includes several IP opportunities:

- AI Grading Algorithm – Proprietary multi-sensor fusion model
- System Architecture – Integrated IoT + AI design

7. BUDGET AND JUSTIFICATION

Table 4 - Budget and Justification Table

Item	Description	Quantity	Unit Price (LKR)
Raspberry Pi 4 Model B (4GB) – common	Edge computing device for AI inference, image processing, and local data aggregation	1	30,000.00
ESP32 Development Board	Microcontroller for sensor interfacing and IoT communication	1	2,000.00
High-Resolution Camera Module (8MP/12MP)	Capturing dried fish images for defect detection	1	6,000.00
Moisture Sensor (Food-Grade Type)	Measuring residual moisture percentage in dried fish	1	7,500.00
VOC Gas Sensor (MQ-135 / Industrial Type)	Detecting odor variations and spoilage indicators	1	4,000.00
LED Controlled Lighting Setup	Stable lighting for consistent image capture	1	5,000.00
Load Cell + HX711 Module	Measuring fish batch weight for quality assessment	1	3,000.00
Protective Industrial Enclosure	Protecting electronic components in processing environment	1	6,000.00
Power Supply & Wiring	Stable power distribution for system components	1	3,000.00
Mounting Frame & Camera Stand	Mechanical support for camera positioning	1	4,000.00

8. WORK BREAKDOWN STRUCTURE (WBS)

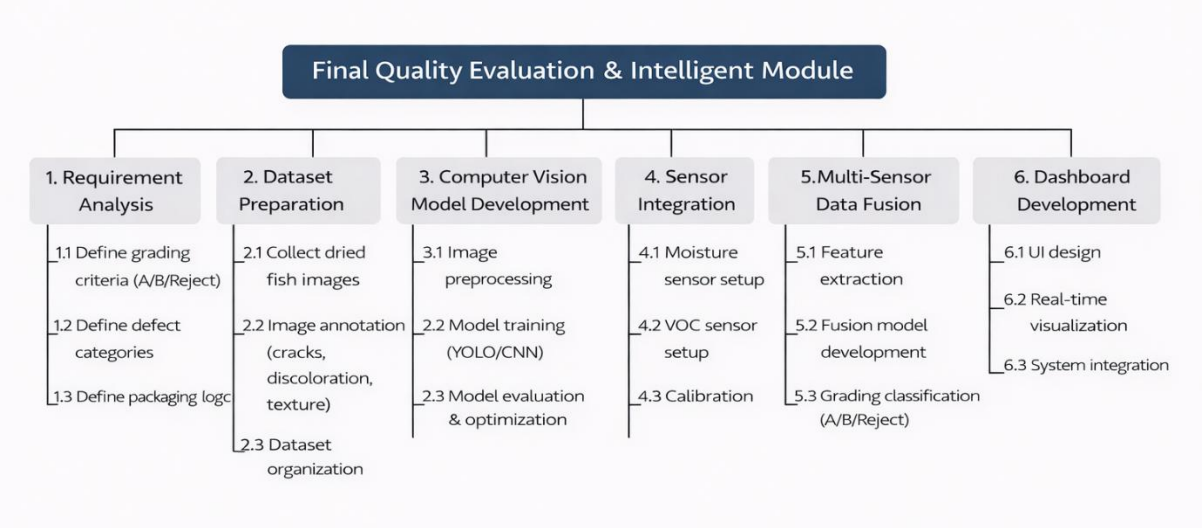


Figure 2 - Work Breakdown Structure

9. GANTT CHART

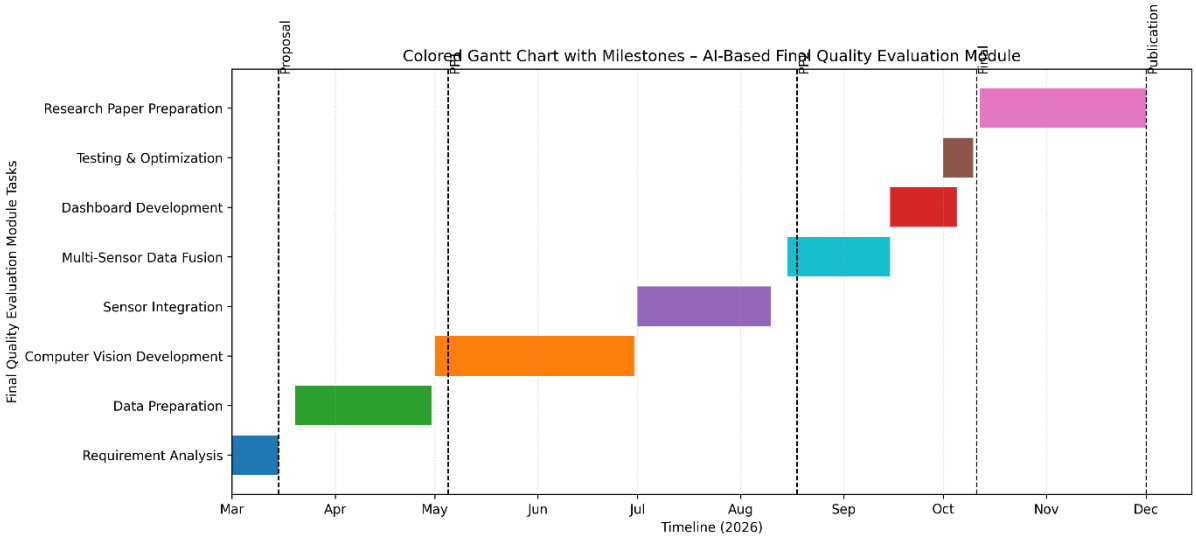


Figure 3 - Gant Chart

10. REFERENCE LIST

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11. APPENDICES

Appendix A – Survey / Interview Instruments

To understand current practices in Maldive fish processing and validate the need for an automated system, brief interviews or surveys may be conducted with fish processing operators and quality control personnel. These instruments aim to capture real-world challenges in grading and the potential benefits of AI-based inspection systems.Sample Survey Questions

Example Questions:

1. How is final-stage fish quality currently inspected at your facility?
2. What challenges do you face with manual fish grading during inspection?
3. How much time does the manual inspection process take per batch?
4. What types of defects (e.g., cracks, discoloration) do you commonly observe in the dried fish?
5. Would an AI-based automated inspection system improve your grading process? If yes, how?
6. How do moisture and odor impact your grading decisions, and how would sensors help?

The survey responses will assist in identifying current operational bottlenecks and help validate the need for an AI-based, automated quality evaluation system for dried Maldive fish.

Appendix B – Preliminary User Interface Mockups

he monitoring dashboard will provide a real-time interface for operators to observe system performance and review fish quality results. The UI design will ensure simplicity, ease of use, and non-technical operation suitable for personnel in fish processing environments.

Key UI Features:

- Real-Time Camera Feed: Display of live camera feed of fish inspection.
- Grading Results: Grade A, Grade B, or Reject classification of fish based on defect detection.
- Defect Detection Alerts: Notification system for damaged or low-quality fish.
- System Performance Statistics: Monitoring of system performance (e.g., processing speed, accuracy).
- Quality Analytics Reports: Batch-specific quality reports with trend analysis for improved decision-making.

Appendix C – Risk Analysis

The project may face several technical and operational risks during the development and deployment stages. These risks are addressed with mitigation strategies to ensure the system's smooth implementation.

Risk	Description	Mitigation Strategy
Insufficient Training Data	Limited dataset of dried fish images may affect model accuracy.	Collect images from diverse sources and apply data augmentation techniques to expand the dataset.
Lighting Variations	Poor lighting conditions could affect the quality of image data captured by cameras.	Use controlled LED lighting near the camera to ensure consistent image quality during inspection.
AI Model Overfitting	The model may perform well on training data but poorly on new, unseen data.	Use a validation dataset, perform cross-validation, and employ regularization techniques to reduce overfitting.
Hardware Integration Issues	Challenges in integrating the camera system with the existing conveyor system could delay the project.	Conduct incremental testing and debugging to address integration issues early in development.
Real-Time Processing Delay	The AI inference process may cause delays in real-time conveyor operations.	Optimize the model for faster inference and use edge computing hardware to process data locally, reducing latency.

Appendix D – Ethical Considerations

This research does not involve human subjects, personal data, or medical experimentation. The study focuses on industrial seafood processing and the development of an AI-based quality inspection system. Despite this, the project adheres to the following ethical guidelines to ensure responsible use of data and technology.

Ethical Guidelines:

- Data Use: All data collected will be used solely for academic research purposes and will not be shared with unauthorized third parties.
- Collaboration with Processing Facilities: Any collaboration with fish processing facilities will be conducted with full organizational consent and clear agreements on data usage.

- Data Privacy: No personal or sensitive information will be collected through the system, as the focus is on industrial product inspection.
- Data Security: All data collected, including images and sensor readings, will be securely stored in accordance with industry-standard cybersecurity practices.